ImMAP-SB: MAP estimation on the bridge,
and I^{2}SB as one of its step-size choices

ImMAP project notes

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Abstract

We derive a family of bridge samplers from one optimization problem: the MAP estimate of the target $\boldsymbol{x}_0$ given the bridge's prior endpoint $\boldsymbol{x}_1$ and, in reconstruction, the measurement $\boldsymbol{y}$. Three steps, each proved below, lead from the objective to the algorithm. *(i) Smoothing.* The bridge supplies, at every time t, a Gaussian-smoothed version $J_\sigma = \log p_\sigma(\cdot \mid \boldsymbol{x}_1)$ of the log-posterior. It is concave for large σ, and its maximisers converge to the MAP estimate as $\sigma \to 0$. So MAP is approached by continuation. *(ii) The gradient step.* By Tweedie's formula the trained regressor gives ∇J_σ exactly, and the curvature of J_σ is bounded by σ^{-2}, so every step of size $h\sigma^2$, $h \in (0, 2)$, provably increases J_σ. *(iii) Staying where the gradient is known.* The network has only seen bridge states, where $\|\nabla J_\sigma\|$ concentrates away from zero, not the stationary points the ascent heads for. Requiring every iterate to be a bridge state at the next noise level fixes the noise that must be injected after each step. The result is a two-parameter family of steps. I^{2}SB is the member whose step size equals the variance removed, $h\sigma_t^2 = \sigma_t^2 - \sigma_s^2$; it is the only step whose deterministic part is the exact conditional mean of the next bridge state. The probability-flow (DDIM-type) step and fresh resampling are the other natural members. Measurement data enter as the same ascent on the posterior-smoothed objective. Under a Gaussian approximation, its gradient is the CG data-consistency step of I^{2}SB-Recon, with a weight λ that should follow the noise level rather than stay fixed.

Notation and conventions. $\boldsymbol{x}_0$ is the target (the fully sampled image), $\boldsymbol{x}_1$ the bridge's prior endpoint (in MRI: the zero-filled image $\boldsymbol{E}^H\boldsymbol{y}$, the ACS low-pass image, or a guide contrast), $t \in [0, 1]$ the bridge time with $t = 0$ at $\boldsymbol{x}_0$. The analysis is written for real vectors in $\mathbb{R}^n$ with $\mathcal{N}(\boldsymbol{0}, \sigma^2\boldsymbol{I})$ noise. A complex image in $\mathbb{C}^N$ with $\mathcal{CN}(\boldsymbol{0}, \sigma^2\boldsymbol{I})$ noise is the real case in $\mathbb{R}^{2N}$ with variance $\sigma^2/2$ per component. With the Wirtinger gradient $\partial/\partial\bar{\boldsymbol{u}}$ every update formula below is unchanged (the factors of 2 cancel); only the constant in the concavity threshold of Proposition 3 moves by $\sqrt{2}$.

1 The bridge is a ladder of denoising problems

The Brownian bridge between $\boldsymbol{x}_0$ and $\boldsymbol{x}_1$ has marginals

$$\boldsymbol{x}_t = (1-t)\,\boldsymbol{x}_0 + t\,\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(t)\,\boldsymbol{\varepsilon}, \qquad \sigma_{\mathrm{sb}}(t)^2 = V\,t(1-t), \qquad \boldsymbol{\varepsilon} \perp (\boldsymbol{x}_0, \boldsymbol{x}_1), \tag{1}$$

with total variance $V = 4\tau^2$ ($\max_t \sigma_{\mathrm{sb}} = \tau$). I^{2}SB [1] trains a regressor with the squared loss $\mathbb{E}\|D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) - \boldsymbol{x}_0\|^2$ over pairs, times and bridge noise. Its minimiser is the conditional mean

$$D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1]. \tag{2}$$

Lemma 1 (Debiased state)**.** *Fix $\boldsymbol{x}_1$ and set $\boldsymbol{u}_t \coloneqq (\boldsymbol{x}_t - t\,\boldsymbol{x}_1)/(1-t)$ for $t < 1$. Then*

$$\boldsymbol{u}_t = \boldsymbol{x}_0 + \sigma(t)\,\boldsymbol{\varepsilon}, \qquad \sigma(t)^2 = \frac{Vt}{1-t}, \qquad t = \frac{\sigma^2}{\sigma^2 + V}, \tag{3}$$

and $t \mapsto \sigma(t)$ is an increasing bijection $[0,1) \to [0,\infty)$. As a process, $\sigma^2 \mapsto \boldsymbol{u}$ is $\boldsymbol{x}_0$ plus a Brownian motion run in the variance σ^2: $\boldsymbol{u} = \boldsymbol{x}_0 + \boldsymbol{W}(\sigma^2)$, $\boldsymbol{W}(0) = \boldsymbol{0}$, independent of $(\boldsymbol{x}_0, \boldsymbol{x}_1)$.

Proof. The marginal statement is (1) divided by $1-t$. For the process, write the bridge as $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1 + \sqrt{V}\,\boldsymbol{B}_t$ with $\boldsymbol{B}$ a standard Brownian bridge on $[0,1]$, and use Doob's representation $\boldsymbol{B}_t = (1-t)\,\boldsymbol{W}_0(t/(1-t))$ with $\boldsymbol{W}_0$ a standard Brownian motion. Then $\boldsymbol{u}_t = \boldsymbol{x}_0 + \sqrt{V}\,\boldsymbol{W}_0(t/(1-t)) = \boldsymbol{x}_0 + \boldsymbol{W}(Vt/(1-t))$ with $\boldsymbol{W}(s) = \sqrt{V}\,\boldsymbol{W}_0(s/V)$, again a standard Brownian motion. □

Since $(\boldsymbol{x}_t, \boldsymbol{x}_1) \mapsto (\boldsymbol{u}_t, \boldsymbol{x}_1)$ is invertible, the regressor is the conditional MMSE denoiser at level $\sigma(t)$,

$$D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}_t, \boldsymbol{x}_1]. \tag{4}$$

One network therefore denoises $\boldsymbol{x}_0$ at every noise level, given $\boldsymbol{x}_1$. The rest of this note turns that into a MAP solver.

Remark 1 (General schedules). For an I^{2}SB schedule, $\boldsymbol{x}_t = \mu_0(t)\boldsymbol{x}_0 + \mu_1(t)\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(t)\boldsymbol{\varepsilon}$ with $\mu_0 + \mu_1 = 1$, the bridge is a Brownian bridge under a time change. Everything below holds with $\boldsymbol{u}_t = (\boldsymbol{x}_t - \mu_1\boldsymbol{x}_1)/\mu_0$ and $\sigma = \sigma_{\mathrm{sb}}/\mu_0$.

2 The objective, and its smoothing

Given $\boldsymbol{x}_1$ we want

$$\hat{\boldsymbol{x}}_{\mathrm{MAP}} = \arg\max_{\boldsymbol{x}} \log p(\boldsymbol{x} \mid \boldsymbol{x}_1), \tag{5}$$

and in reconstruction $\arg\max_{\boldsymbol{x}} \log p(\boldsymbol{x} \mid \boldsymbol{x}_1, \boldsymbol{y})$ (Section 8). The conditional prior has no closed form and is in general non-concave. We follow the classical route for such problems, graduated optimisation: replace the objective by a smoothed one, solve the smooth problem, and reduce the smoothing. The smoothing that the bridge provides is Gaussian. Let φ_σ be the $\mathcal{N}(\boldsymbol{0}, \sigma^2\boldsymbol{I})$ density and

$$p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1) = \int \varphi_\sigma(\boldsymbol{u} - \boldsymbol{x})\, p(\boldsymbol{x} \mid \boldsymbol{x}_1)\, d\boldsymbol{x}, \qquad J_\sigma(\boldsymbol{u}) \coloneqq \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1), \tag{6}$$

the law of $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$ given $\boldsymbol{x}_1$, i.e. the law of the debiased bridge state at time $t(\sigma)$. Three facts make J_σ a good surrogate.

Proposition 2 (Tweedie / Miyasawa)**.** $\nabla J_\sigma(\boldsymbol{u}) = \dfrac{\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] - \boldsymbol{u}}{\sigma^2} = \dfrac{D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) - \boldsymbol{u}}{\sigma^2}$ *at* $\sigma = \sigma(t)$.

Proof. $\nabla_{\boldsymbol{u}}\varphi_\sigma(\boldsymbol{u} - \boldsymbol{x}) = \sigma^{-2}(\boldsymbol{x} - \boldsymbol{u})\varphi_\sigma(\boldsymbol{u} - \boldsymbol{x})$. Differentiate (6) under the integral, $\sigma^2 \nabla p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1) = \int (\boldsymbol{x} - \boldsymbol{u})\varphi_\sigma(\boldsymbol{u} - \boldsymbol{x}) p(\boldsymbol{x} \mid \boldsymbol{x}_1)\, d\boldsymbol{x} = p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1)(\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] - \boldsymbol{u})$, divide by p_σ, and use (4). □

So the network does not approximate the objective, it computes its gradient. The residual $D_\theta - \boldsymbol{u}$ is $\sigma^2 \nabla J_\sigma$.

Proposition 3 (Curvature)**.**

$$\nabla^2 J_\sigma(\boldsymbol{u}) = \frac{\operatorname{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] - \sigma^2 \boldsymbol{I}}{\sigma^4}. \tag{7}$$

Consequently (a) $-\nabla^2 J_\sigma \preceq \sigma^{-2}\boldsymbol{I}$ *everywhere, and* (b) *if* $p(\cdot \mid \boldsymbol{x}_1)$ *is supported in a ball of radius* R*, then* J_σ *is concave for every* $\sigma \geq R$*.*

Proof. By Bayes, $\nabla_{\boldsymbol{u}} \log p(\boldsymbol{x} \mid \boldsymbol{u}, \boldsymbol{x}_1) = \sigma^{-2}(\boldsymbol{x} - \boldsymbol{u}) - \nabla J_\sigma(\boldsymbol{u}) = \sigma^{-2}(\boldsymbol{x} - \boldsymbol{m})$ with $\boldsymbol{m} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$, by Proposition 2. Hence $\nabla_{\boldsymbol{u}} \boldsymbol{m} = \int \boldsymbol{x}\,(\boldsymbol{x} - \boldsymbol{m})^\top \sigma^{-2} p(\boldsymbol{x} \mid \boldsymbol{u}, \boldsymbol{x}_1)\, d\boldsymbol{x} = \sigma^{-2} \operatorname{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$, and differentiating $\nabla J_\sigma = \sigma^{-2}(\boldsymbol{m} - \boldsymbol{u})$ gives (7). (a) holds because a covariance is positive semidefinite. For (b), any law on a ball of radius R around $\boldsymbol{c}$ has $\boldsymbol{a}^\top \operatorname{Cov} \boldsymbol{a} \leq \mathbb{E}[(\boldsymbol{a}^\top(\boldsymbol{x} - \boldsymbol{c}))^2] \leq R^2 \|\boldsymbol{a}\|^2$, so $\operatorname{Cov} \preceq R^2 \boldsymbol{I} \preceq \sigma^2 \boldsymbol{I}$. □

At large σ the smoothed problem is concave and has a single maximiser; images have bounded intensities, so (b) applies with R of the order of the image norm. It remains to show that reducing σ leads back to the problem we want.

Assumption 1. $p(\cdot \mid \boldsymbol{x}_1)$ is continuous on $\mathbb{R}^n$, vanishes outside a compact set, and has a unique maximiser $\boldsymbol{x}^\star$.

Proposition 4 (Smoothed MAP → MAP)**.** *Under Assumption 1,* J_σ *attains its maximum for every* $\sigma > 0$*, and every maximiser* $\boldsymbol{u}_\sigma^\star$ *satisfies* $\boldsymbol{u}_\sigma^\star \to \boldsymbol{x}^\star$ *as* $\sigma \to 0$*.*

Proof. Write $p = p(\cdot \mid \boldsymbol{x}_1)$. It is bounded and uniformly continuous, with modulus ω. For any $\delta > 0$, $|p_\sigma(\boldsymbol{u}) - p(\boldsymbol{u})| \leq \mathbb{E}|p(\boldsymbol{u} - \sigma\boldsymbol{\varepsilon}) - p(\boldsymbol{u})| \leq \omega(\delta) + 2\|p\|_\infty \Pr(\sigma\|\boldsymbol{\varepsilon}\| > \delta)$, so $p_\sigma \to p$ uniformly on $\mathbb{R}^n$. Also p_σ is continuous and tends to 0 at infinity, so its maximum is attained. Let B be an open ball around $\boldsymbol{x}^\star$. By continuity, compact support and uniqueness, $m \coloneqq \sup_{\boldsymbol{u} \notin B} p(\boldsymbol{u}) < p(\boldsymbol{x}^\star)$. Choose σ small enough that $\sup |p_\sigma - p| < \frac{1}{2}(p(\boldsymbol{x}^\star) - m)$. Then for $\boldsymbol{u} \notin B$, $p_\sigma(\boldsymbol{u}) < m + \frac{1}{2}(p(\boldsymbol{x}^\star) - m) < p_\sigma(\boldsymbol{x}^\star)$, so every maximiser of p_σ lies in B. □

Propositions 3(b) and 4 are the two ends of a continuation: an easy concave problem at large σ, the MAP problem in the limit $\sigma \to 0$, and in between a family that the bridge indexes by t. A local method that tracks a maximiser while σ decreases solves (5) when the tracked maximiser is the global one. That is guaranteed at the concave end and is the usual heuristic of graduated optimisation in between.

3 The gradient step

Gradient ascent on J_σ with step size $h\sigma^2$ reads, by Proposition 2,

$$\boldsymbol{u}^+ = \boldsymbol{u} + h\sigma^2 \nabla J_\sigma(\boldsymbol{u}) = (1 - h)\,\boldsymbol{u} + h\,D_\theta(\boldsymbol{u}), \tag{8}$$

a move of the fraction h of the way toward the denoiser output. The curvature bound makes σ^2 the natural unit of step size.

Proposition 5 (Ascent lemma)**.** *For* $h \in (0, 2)$*,*

$$J_\sigma(\boldsymbol{u}^+) \;\geq\; J_\sigma(\boldsymbol{u}) + h\left(1 - \tfrac{h}{2}\right)\sigma^2 \|\nabla J_\sigma(\boldsymbol{u})\|^2 \;=\; J_\sigma(\boldsymbol{u}) + h\left(1 - \tfrac{h}{2}\right)\frac{\|D_\theta(\boldsymbol{u}) - \boldsymbol{u}\|^2}{\sigma^2}.$$

The guaranteed increase is largest at $h = 1$*, the full jump to* $D_\theta(\boldsymbol{u})$*.*

Proof. $f = -J_\sigma$ has $\nabla^2 f \preceq L\boldsymbol{I}$ with $L = \sigma^{-2}$ (Proposition 3(a)). The descent lemma for such f, $f(\boldsymbol{u} - \alpha\nabla f) \le f(\boldsymbol{u}) - \alpha(1 - \frac{L\alpha}{2})\|\nabla f\|^2$, follows from Taylor's theorem with that bound along the segment. Put $\alpha = h\sigma^2$. □

So every step (8) with $h \in (0, 2)$ is a certified ascent step, without any assumption on the shape of the prior; a one-shot reconstruction network ($h = 1$, a single denoiser call) is the single optimal step at the noise level it was trained for. The fixed points of (8) are the points with $D_\theta(\boldsymbol{u}) = \boldsymbol{u}$, the stationary points of J_σ.

4 Staying where the gradient is known

The gradient in Proposition 2 is exact for the true conditional mean. The network approximates it only where it was trained: at bridge states, i.e. at points $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$ distributed as $p_\sigma(\cdot \mid \boldsymbol{x}_1)$. An ascent at a fixed level heads for exactly the points that are not of this kind.

Lemma 6 (Bridge states have a large residual)**.** *For $\boldsymbol{u} \sim p_\sigma(\cdot \mid \boldsymbol{x}_1)$,*

$$\mathbb{E}\|D_\theta(\boldsymbol{u}) - \boldsymbol{u}\|^2 = \sigma^4\,\mathbb{E}\|\nabla J_\sigma(\boldsymbol{u})\|^2 = n\sigma^2 - \mathrm{mmse}(\sigma), \qquad \mathrm{mmse}(\sigma) \coloneqq \mathbb{E}\|D_\theta(\boldsymbol{u}) - \boldsymbol{x}_0\|^2 \le n\sigma^2. \tag{9}$$

Proof. $\boldsymbol{u} - \boldsymbol{x}_0 = (\boldsymbol{u} - D_\theta) + (D_\theta - \boldsymbol{x}_0)$, and the two terms are orthogonal in expectation because the estimation error of a conditional mean is orthogonal to every function of the data, here $\boldsymbol{u} - D_\theta(\boldsymbol{u})$. So $n\sigma^2 = \mathbb{E}\|\boldsymbol{u} - D_\theta\|^2 + \mathrm{mmse}$. The bound is the error of the trivial estimator $D_\theta(\boldsymbol{u}) = \boldsymbol{u}$. □

For images, $\mathrm{mmse}(\sigma) \ll n\sigma^2$ over most of the ladder: the denoiser removes most of the noise (for data near a d-dimensional set, $\mathrm{mmse} \approx d\sigma^2$ at small σ). So on the training distribution the residual concentrates at $\|D_\theta - \boldsymbol{u}\| \approx \sigma\sqrt{n - \mathrm{mmse}/\sigma^2}$, far from zero, whereas at the points the ascent converges to it is zero. A maximiser of J_σ is a mode of p_σ, and in high dimension the mode is not a typical point. Running (8) to convergence at one level, or shrinking σ more slowly than the steps remove noise, therefore evaluates the network ever further from its training data. We impose the opposite:

Requirement (R). *Every iterate at level σ_k is a bridge state: $\boldsymbol{u}_k - \boldsymbol{x}_0 \sim \mathcal{N}(\boldsymbol{0}, \sigma_k^2\boldsymbol{I})$, independent of $\boldsymbol{x}_0$.*

With an accurate denoiser, the step (8) leaves noise of level $(1 - h)\sigma$ in the iterate, less than the level σ it came from. Requirement (R) at the next level σ' decides how much fresh noise must be added back.

Proposition 7 (Noise injection)**.** *Let $\boldsymbol{u}$ satisfy (R) at level σ, and take the step $\boldsymbol{u}' = (1 - h)\,\boldsymbol{u} + h\,D_\theta(\boldsymbol{u}) + \gamma\,\boldsymbol{z}$ with $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ fresh. With an exact denoiser of the noise ($D_\theta(\boldsymbol{u}) = \boldsymbol{x}_0$), $\boldsymbol{u}'$ satisfies (R) at level σ' if and only if*

$$\gamma^2 = \sigma'^2 - (1 - h)^2\sigma^2, \qquad \textit{which requires} \qquad (1 - h)\,\sigma \le \sigma'. \tag{10}$$

Proof. $\boldsymbol{u}' - \boldsymbol{x}_0 = (1 - h)\sigma\boldsymbol{\varepsilon} + \gamma\boldsymbol{z}$ is Gaussian with variance $(1 - h)^2\sigma^2 + \gamma^2$ and independent of $\boldsymbol{x}_0$. □

A step is therefore described by two numbers: the step size h and the next level σ'. Kadkhodaie and Simoncelli [2] parameterise the second as $\sigma' = (1 - \beta h)\,\sigma$, $\beta \in [0, 1]$; the constraint in (10) is $\beta \le 1$, and $\beta = 1$ injects nothing. The oracle assumption $D_\theta(\boldsymbol{u}) = \boldsymbol{x}_0$ is what makes (R) checkable. The next result removes it and identifies the step size at which the deterministic part of the step is right for the true conditional mean.

Proposition 8 (The exact reverse step)**.** *For $\sigma_s < \sigma_t$ and $r := \sigma_s/\sigma_t$,*

$$\mathbb{E}[\boldsymbol{u}_s \mid \boldsymbol{u}_t, \boldsymbol{x}_1] = r^2\, \boldsymbol{u}_t + (1 - r^2)\, D_\theta(\boldsymbol{u}_t), \tag{11}$$

$$\mathrm{Cov}[\boldsymbol{u}_s \mid \boldsymbol{u}_t, \boldsymbol{x}_1] = \sigma_s^2(1 - r^2)\, \boldsymbol{I} + (1 - r^2)^2\, \mathrm{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}_t, \boldsymbol{x}_1]. \tag{12}$$

Hence the gradient step (8) has the exact conditional mean of the next state if and only if

$$h = 1 - r^2, \qquad \textit{i.e.} \qquad h\,\sigma_t^2 = \sigma_t^2 - \sigma_s^2, \tag{13}$$

the step size equals the variance removed. With this h, the injection (10) is $\gamma^2 = \sigma_s^2(1 - r^2)$, the first term of (12).

Proof. By Lemma 1, given $\boldsymbol{x}_0$ the pair $(\boldsymbol{u}_s, \boldsymbol{u}_t)$ is a Brownian motion observed at variances $\sigma_s^2 < \sigma_t^2$. Conditioning a Brownian motion on its value at a later time gives a Brownian bridge: $\boldsymbol{u}_s \mid \boldsymbol{u}_t, \boldsymbol{x}_0 \sim \mathcal{N}(\boldsymbol{x}_0 + r^2(\boldsymbol{u}_t - \boldsymbol{x}_0),\ \sigma_s^2(1 - r^2)\boldsymbol{I})$. The Markov property makes $\boldsymbol{x}_1$ (and any noisier state) irrelevant once $\boldsymbol{u}_t$ and $\boldsymbol{x}_0$ are given. Averaging over $\boldsymbol{x}_0 \sim p(\cdot \mid \boldsymbol{u}_t, \boldsymbol{x}_1)$ with the laws of total expectation and total covariance gives (11) and (12). The mean of (8) is $(1 - h)\boldsymbol{u}_t + hD_\theta$, which equals (11) for $D_\theta \neq \boldsymbol{u}_t$ iff $h = 1 - r^2$. Finally $\sigma_s^2 - r^4\sigma_t^2 = \sigma_s^2(1 - r^2)$. □

The step (13) plus the injection (10) has the right mean and drops only the term $(1-r^2)^2\,\mathrm{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}_t, \boldsymbol{x}_1]$ of the covariance: the spread of the endpoint around its estimate, which a single point estimate cannot represent. That term is $O((\sigma_t^2 - \sigma_s^2)^2)$, so it vanishes on fine ladders.

5 The step on the bridge

Forming $\boldsymbol{u}$ explicitly is unstable near $t = 1$ (the factor $1/(1 - t)$), so the step is carried out in bridge coordinates. The iterate is $(\boldsymbol{x}_k, t_k)$, with $\sigma_k = \sigma(t_k)$ and $\boldsymbol{v}_k = D_\theta(\boldsymbol{x}_k, t_k, \boldsymbol{x}_1)$; the next level σ' gives $t' = \sigma'^2/(\sigma'^2 + V)$; the stepped point $(1 - h)\boldsymbol{u}_k + h\boldsymbol{v}_k$ is mapped onto the bridge at t' and the injection is scaled by $1 - t'$.

Proposition 9 (Bridge coordinates)**.** *The step of Proposition 7, mapped to $\boldsymbol{x}$, is*

$$\boldsymbol{x}_{k+1} = c_x\,\boldsymbol{x}_k + c_v\,\boldsymbol{v}_k + c_1\,\boldsymbol{x}_1 + \gamma_k\,\boldsymbol{\varepsilon}_k, \qquad c_x = \frac{(1 - h)(1 - t')}{1 - t_k}, \quad c_v = h(1 - t'), \quad c_1 = t' - c_x t_k, \tag{14}$$

with $c_x + c_v + c_1 = 1$ and the injection $\gamma_k = (1 - t')\gamma$ (γ from (10)), i.e.

$$\gamma_k^2 = \sigma_{\mathrm{sb}}(t')^2\,(1 - \rho^2), \qquad \rho = \frac{(1 - h)\,\sigma_k}{\sigma'} = \frac{1 - h}{1 - \beta h} \in [0, 1]. \tag{15}$$

With an exact denoiser, $\boldsymbol{x}_{k+1}$ is distributed exactly as the bridge marginal (1) at t'.

Proof. $\boldsymbol{x}_{k+1} = (1 - t')\big[(1 - h)\frac{\boldsymbol{x}_k - t_k\boldsymbol{x}_1}{1 - t_k} + h\boldsymbol{v}_k\big] + t'\boldsymbol{x}_1 + (1 - t')\gamma\boldsymbol{z}$ gives the coefficients, which sum to $(1-h)(1-t')+h(1-t')+t' = 1$. The noise carried from $\boldsymbol{x}_k$ is $c_x\sigma_{\mathrm{sb}}(t_k)\boldsymbol{\varepsilon} = (1-h)(1-t')\sigma_k\boldsymbol{\varepsilon} = \rho\,\sigma_{\mathrm{sb}}(t')\boldsymbol{\varepsilon}$, and $(1 - t')^2\gamma^2 = (1 - t')^2(\sigma'^2 - (1 - h)^2\sigma_k^2) = \sigma_{\mathrm{sb}}(t')^2(1 - \rho^2)$. □

This is `sb.immap_sb_ascent.bridge_ascent_step`.

6 I^{2}SB and its neighbours are step-size choices

The derivation leaves the step size h and the next level σ' free. Prescribing the next level by a fixed time ladder, $\sigma' = \sigma(s)$ for a step $t \to s$, and writing $r = \sigma(s)/\sigma(t)$, three choices of h stand out.

Proposition 10 (I^{2}SB)**.** *With $h = 1 - r^2$ (the exact-mean step (13)), Proposition 9 is exactly I^{2}SB's reverse step, the bridge posterior with $\boldsymbol{x}_0$ replaced by $\boldsymbol{v} = D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1)$:*

$$\boldsymbol{x}_s \sim \mathcal{N}\Big((1 - \tfrac{s}{t})\boldsymbol{v} + \tfrac{s}{t}\,\boldsymbol{x}_t,\; V\,\tfrac{s(t-s)}{t}\,\boldsymbol{I}\Big). \tag{16}$$

In Kadkhodaie–Simoncelli form, $\beta = 1/(1 + r)$.

Proof. With $\sigma^2 = Vt/(1-t)$, $r^2 = s(1-t)/(t(1-s))$, so $h = (t-s)/(t(1-s))$. Then $c_v = h(1-s) = 1 - \frac{s}{t}$, $c_x = r^2(1-s)/(1-t) = \frac{s}{t}$ and $c_1 = s - \frac{s}{t}t = 0$, the mean of (16). $\rho = (1-h)\sigma(t)/\sigma(s) = r$, so $\gamma^2 = \sigma_{\mathrm{sb}}(s)^2(1-r^2) = Vs(1-s)\frac{t-s}{t(1-s)} = V\frac{s(t-s)}{t}$. Finally $1 - \beta h = r$ gives $\beta = (1-r)/h = 1/(1+r)$. □

This is `sb.base.reverse_sample` with `posterior="ddpm"`; `tests/test_immap_sb_ascent.py` checks the identity numerically.

Corollary 11 (The other natural members)**.** *1. **No injected noise**, $\gamma = 0$: $h = 1 - r$ ($\beta = 1$). This is the explicit Euler step in σ of the probability-flow ODE $d\boldsymbol{u}/d\sigma = (\boldsymbol{u} - D_\theta(\boldsymbol{u}))/\sigma = -\sigma\nabla J_\sigma$ [3], and the largest noise reduction per unit of step size that still satisfies (R).*

*2. **Fresh resampling**, $h = 1$: jump to $\boldsymbol{v}$ and draw a new bridge state around it, $\boldsymbol{x}_s = (1-s)\boldsymbol{v} + s\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(s)\boldsymbol{z}$ ($\rho = 0$; `posterior="interpolant"`).*

*3. **ImMAP** [2]: h_k from its own schedule (growing toward 1), β free, and σ' either open-loop or estimated from the residual through Lemma 6, $\hat{\sigma}^2 \approx \|\boldsymbol{v} - \boldsymbol{u}\|^2/n$ when mmse $\ll n\sigma^2$ (`sigma_mode`).*

Proof. (1) $\gamma = 0$ in (10) is $(1-h)\sigma(t) = \sigma(s)$. The Euler step from σ_t to σ_s is $\boldsymbol{u} + (\sigma_s - \sigma_t)(\boldsymbol{u} - D_\theta)/\sigma_t = r\boldsymbol{u} + (1-r)D_\theta$, the step (8) with $h = 1 - r$. (2) and (3) are (14) with $h = 1$, and with the stated schedules. □

Remark 2 (Dropping the noise from I^{2}SB's step). Keeping $h = 1 - r^2$ but setting $\gamma = 0$ (`deterministic=True`) violates (10): with an exact denoiser the next state has noise level $r^2\sigma(t) = r\,\sigma(s) < \sigma(s)$, and the next network call is told it is at $\sigma(s)$. The consistent deterministic sampler is (1) above, $h = 1 - r$.

Remark 3 (Euler–Maruyama). The reverse-time SDE in variance time, $d\boldsymbol{u} = -\nabla J_\sigma\, d(\sigma^2) + d\boldsymbol{W}$, has the Euler–Maruyama step $\boldsymbol{u}_s = \boldsymbol{u}_t + (\sigma_t^2 - \sigma_s^2)\nabla J_{\sigma_t}(\boldsymbol{u}_t) + \sqrt{\sigma_t^2 - \sigma_s^2}\,\boldsymbol{z}$. Its drift is I^{2}SB's (13); its noise variance $\sigma_t^2 - \sigma_s^2$ exceeds I^{2}SB's $\sigma_s^2(1-r^2) = r^2(\sigma_t^2 - \sigma_s^2)$, which is the exact conditional variance (12) given $\boldsymbol{x}_0$.

7 What the step size does: averaging the gradient steps

The exact-mean step size (13) has a simple cumulative effect.

member	step size h	next level σ'	injected γ^2 (in $\boldsymbol{u}$)
I^{2}SB (`ddpm`)	$1-r^2$	$\sigma(s)$	$\sigma(s)^2(1-r^2)$
probability flow / Euler	$1-r$	$\sigma(s)$	0
fresh resampling (`interpolant`)	1	$\sigma(s)$	$\sigma(s)^2$
ImMAP / KS	schedule h_k	$(1-\beta h)\sigma$ or $\hat{\sigma}$	$\sigma'^2-(1-h)^2\sigma^2$

Table 1: One step $\boldsymbol{u}' = (1-h)\boldsymbol{u} + h\boldsymbol{v} + \gamma\boldsymbol{z}$ of gradient ascent on J_σ with the injection required by (R); $r = \sigma(s)/\sigma(t)$.

Proposition 12 (Precision-weighted average)**.** *Run I^{2}SB's step down a ladder $\sigma_0 = \infty > \sigma_1 > \cdots > \sigma_K$, with denoiser outputs $\boldsymbol{v}^{(k)}$ at level σ_k. Then $\boldsymbol{u}_K = \bar{\boldsymbol{u}}_K + \boldsymbol{m}_K$, with $\boldsymbol{m}_K$ the sampler's own noise, of variance σ_K^2 per component, and*

$$\bar{\boldsymbol{u}}_K = \sigma_K^2 \sum_{k=0}^{K-1} \Big(\frac{1}{\sigma_{k+1}^2} - \frac{1}{\sigma_k^2}\Big)\, \boldsymbol{v}^{(k)}, \tag{17}$$

a convex combination: the weights are nonnegative and sum to one.

Proof. Split $\boldsymbol{u}_k = \bar{\boldsymbol{u}}_k + \boldsymbol{m}_k$ into the part driven by the $\boldsymbol{v}$'s and the part driven by the injected noise. Dividing the $\bar{\boldsymbol{u}}$ part of $\boldsymbol{u}_{k+1} = r_k^2\boldsymbol{u}_k + (1-r_k^2)\boldsymbol{v}^{(k)} + \gamma_k\boldsymbol{z}_k$ by σ_{k+1}^2 and using $r_k^2 = \sigma_{k+1}^2/\sigma_k^2$ gives $\bar{\boldsymbol{u}}_{k+1}/\sigma_{k+1}^2 = \bar{\boldsymbol{u}}_k/\sigma_k^2 + (\sigma_{k+1}^{-2} - \sigma_k^{-2})\boldsymbol{v}^{(k)}$, which telescopes from $1/\sigma_0^2 = 0$. The weights sum to $\sigma_K^2(\sigma_K^{-2} - \sigma_0^{-2}) = 1$, and $\operatorname{Var}\boldsymbol{m}_{k+1} = r_k^4\sigma_k^2 + \sigma_{k+1}^2(1-r_k^2) = \sigma_{k+1}^2$. □

So with h tied to the variance removed, the iterate is a bridge state around a running average of every denoiser output so far, each weighted by the precision σ^{-2} gained when it was made: early, blurry estimates receive little weight, late sharp ones most of it. Fresh resampling ($h = 1$) keeps only the last estimate. The comparison `ddpm` vs. `interpolant` is therefore a direct test of the value of the averaging.

Remark 4 (The estimates sharpen)*.* Along the exact reverse bridge the denoiser outputs $\boldsymbol{v}^{(k)} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}_{t_k}, \boldsymbol{x}_1]$ form a Doob martingale: the bridge is Markov (Lemma 1), so $\boldsymbol{v}^{(k)} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}_{t_k}, \ldots, \boldsymbol{u}_{t_0}, \boldsymbol{x}_1]$, an expectation over a growing history. They run from the posterior mean $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1]$ at $t = 1$ to $\boldsymbol{x}_0$ at $t = 0$. A point-estimate step reproduces this only up to the dropped term of (12).

8 Adding the measurement

In reconstruction the objective is $\arg\max_{\boldsymbol{x}} \log p(\boldsymbol{x} \mid \boldsymbol{x}_1, \boldsymbol{y})$, with $\boldsymbol{y} = \boldsymbol{E}\boldsymbol{x}_0 + \boldsymbol{n}$, $\boldsymbol{n} \sim \mathcal{N}(\boldsymbol{0}, \sigma_n^2\boldsymbol{I})$. Smooth it exactly as before: $J_\sigma^{\boldsymbol{y}}(\boldsymbol{u}) := \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1, \boldsymbol{y})$, the law of $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$ given $(\boldsymbol{x}_1, \boldsymbol{y})$. Propositions 3–8 hold verbatim with $\boldsymbol{x}_1$ replaced by $(\boldsymbol{x}_1, \boldsymbol{y})$, since their proofs use only that $\boldsymbol{\varepsilon}$ is independent of the conditioning variables. In particular:

Proposition 13 (Tweedie with the measurement)**.** $\nabla J_\sigma^{\boldsymbol{y}}(\boldsymbol{u}) = \big(\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}] - \boldsymbol{u}\big)/\sigma^2$.

Proof. $\boldsymbol{\varepsilon}$ is independent of $(\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{y})$, so $p_\sigma(\boldsymbol{u} \mid \boldsymbol{x}_1, \boldsymbol{y}) = \int \varphi_\sigma(\boldsymbol{u} - \boldsymbol{x}) p(\boldsymbol{x} \mid \boldsymbol{x}_1, \boldsymbol{y})\, d\boldsymbol{x}$; repeat the proof of Proposition 2. □

The whole ascent therefore carries over with $\boldsymbol{v}$ replaced by $\bar{\boldsymbol{v}} \approx \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$. The network supplies $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$; the measurement has to be folded in.

Proposition 14 (The data step)**.** *Assume $\boldsymbol{y}$ is conditionally independent of $(\boldsymbol{u}, \boldsymbol{x}_1)$ given $\boldsymbol{x}_0$, and approximate $p(\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1) \approx \mathcal{N}(\boldsymbol{v}, \varsigma^2 \boldsymbol{I})$ around the denoiser output $\boldsymbol{v}$. Then*

$$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}] \approx \bar{\boldsymbol{v}} = \arg\min_{\boldsymbol{x}} \ \|\boldsymbol{y} - \boldsymbol{E}\boldsymbol{x}\|^2 + \lambda \|\boldsymbol{x} - \boldsymbol{v}\|^2 = (\boldsymbol{E}^H \boldsymbol{E} + \lambda \boldsymbol{I})^{-1} (\boldsymbol{E}^H \boldsymbol{y} + \lambda \boldsymbol{v}), \qquad \lambda = \frac{\sigma_n^2}{\varsigma^2}, \tag{18}$$

with equality when the approximation is exact.

Proof. By the assumption, $p(\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{x}_0)\, p(\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1)$. With a Gaussian likelihood and a Gaussian prior the posterior is Gaussian, and its mean is the minimiser of $\|\boldsymbol{y} - \boldsymbol{E}\boldsymbol{x}\|^2/\sigma_n^2 + \|\boldsymbol{x} - \boldsymbol{v}\|^2/\varsigma^2$. □

(18) is the proximal point of the data term at $\boldsymbol{v}$, solved with a few conjugate-gradient iterations warm-started at $\boldsymbol{v}$: the CG step of I^{2}SB-Recon [4] (`sb.i2sb_recon.cg_data_consistency`). The ImMAP-SB step is then a gradient step on the posterior-smoothed objective, prior part by the network and data part by the proximal correction, followed by the injection that (R) requires.

Lemma 15 (How to set λ)**.** *Averaged over bridge states, the posterior variance per component is at most the noise variance: $\frac{1}{n}\mathbb{E}\,\mathrm{tr}\,\mathrm{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] = \mathrm{mmse}(\sigma)/n \le \sigma^2$.*

Proof. $\mathbb{E}\,\mathrm{tr}\,\mathrm{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] = \mathbb{E}\|\boldsymbol{x}_0 - \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]\|^2 = \mathrm{mmse}(\sigma)$, and $\mathrm{mmse}(\sigma) \le n\sigma^2$ by Lemma 6. □

So the approximation of Proposition 14 has $\varsigma^2 \approx c\,\sigma^2$ with $c \in (0, 1]$, and

$$\lambda(\sigma) = \frac{\sigma_n^2}{c\,\sigma^2}. \tag{19}$$

The data weight follows the bridge: at large σ the denoiser's estimate is uncertain and the data dominate; at small σ the estimate is sharp and the proximal step barely moves it. I^{2}SB-Recon fixes $\lambda = 7 \cdot 10^{-5}$, which by (18) amounts to a posterior std $\varsigma = \sigma_n/\sqrt{\lambda}$ that is the same at every noise level. With $\sigma_n > 0$ a small fixed λ fits the measurement noise on the sampled lines at every step. It does so most at the last steps, where the actual posterior std ($\varsigma \lesssim \sigma$, Lemma 15) is far below the implied $\sigma_n/\sqrt{\lambda}$.

Remark 5 (When the measurement is already in $\boldsymbol{x}_1$). The conditional independence in Proposition 14 holds for a guide-contrast start, where $\boldsymbol{x}_1$ is a different scan. For the zero-filled start, $\boldsymbol{x}_1 = \boldsymbol{E}^H \boldsymbol{y}$ is a sufficient statistic for $\boldsymbol{x}_0$ given $\boldsymbol{y}$ (with white noise, the likelihood depends on $\boldsymbol{y}$ only through $\boldsymbol{E}^H \boldsymbol{y}$), so $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$ already: the network sees the data and (18) adds no information in exact arithmetic. It still corrects the network's errors, which is where its value lies in practice. The ACS start is in between: it keeps only the centre lines of $\boldsymbol{y}$, and the outer lines enter through (18) alone.

9 The algorithm

Algorithm 1 ImMAP-SB: annealed gradient ascent on the bridge (`sb/immap_sb_ascent.py`, `immap_sb_ascent`)

Require: endpoint $\boldsymbol{x}_1$; regressor D_θ; total variance V; start level σ_0; step sizes h_k; next levels (open-loop β, a ladder, or estimated); optional measurement $(\boldsymbol{y}, \boldsymbol{E}, \sigma_n)$ and c

$t_0 \leftarrow \sigma_0^2/(\sigma_0^2 + V)$; $\quad \boldsymbol{x} \leftarrow (1 - t_0)\, D_\theta(\boldsymbol{x}_1, 1, \boldsymbol{x}_1) + t_0 \boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(t_0)\, \boldsymbol{\varepsilon}$ ▷ a bridge state at σ_0

for $k = 0, 1, \ldots$ while $\sigma_k > \sigma_{\min}$ **do**

  $\boldsymbol{v} \leftarrow D_\theta(\boldsymbol{x}, t_k, \boldsymbol{x}_1)$ ▷ $\sigma_k^2 \nabla J_{\sigma_k} = \boldsymbol{v} - \boldsymbol{u}$ (Prop. 2)

  if $\boldsymbol{y}$: $\boldsymbol{v} \leftarrow (\boldsymbol{E}^H \boldsymbol{E} + \lambda_k \boldsymbol{I})^{-1}(\boldsymbol{E}^H \boldsymbol{y} + \lambda_k \boldsymbol{v})$, $\lambda_k = \sigma_n^2/(c\,\sigma_k^2)$ ▷ data step (Prop. 14)

  choose h_k and σ' with $(1 - h_k)\sigma_k \le \sigma'$; $\quad t' \leftarrow \sigma'^2/(\sigma'^2 + V)$

  $\boldsymbol{x} \leftarrow c_x \boldsymbol{x} + c_v \boldsymbol{v} + c_1 \boldsymbol{x}_1 + \gamma_k \boldsymbol{\varepsilon}$ (Prop. 9) ▷ gradient step + injection (R)

  $(t_{k+1}, \sigma_{k+1}) \leftarrow (t', \sigma')$

end for

return $\boldsymbol{v}$

The start is a bridge state at a finite σ_0: $\boldsymbol{x}_1$ itself ($\sigma = \infty$) is a fixed point of any bridge-consistent update. The returned estimate is the last denoiser output; for I^{2}SB's ladder ending at $t = 0$, $\sigma_{\mathrm{sb}}(0) = 0$ and the final state equals it.

10 Summary

step of the derivation	statement	where
bridge = denoising ladder	$\boldsymbol{u}_t = \boldsymbol{x}_0 + \sigma(t)\boldsymbol{\varepsilon}$, Brownian in σ^2	Lemma 1
objective	$\max \log p(\boldsymbol{x} \mid \boldsymbol{x}_1[, \boldsymbol{y}])$, smoothed to J_σ	(5), (6)
smoothing is benign	concave for $\sigma \ge R$; maximisers $\to$ MAP as $\sigma \to 0$	Props. 3, 4
gradient	$\sigma^2 \nabla J_\sigma = D_\theta - \boldsymbol{u}$	Prop. 2
step	$\boldsymbol{u}^+ = (1 - h)\boldsymbol{u} + h D_\theta$, ascent for $h \in (0, 2)$	Prop. 5
injection	$\gamma^2 = \sigma'^2 - (1 - h)^2 \sigma^2$, from (R)	Lemma 6, Prop. 7
exact-mean step size	$h\sigma_t^2 = \sigma_t^2 - \sigma_s^2$	Prop. 8
I^{2}SB	that step size, ladder levels	Prop. 10
probability flow, resampling, ImMAP	$h = 1 - r$; $h = 1$; free h_k, β	Cor. 11
I^{2}SB's iterate	precision-weighted average of the $\boldsymbol{v}^{(k)}$	Prop. 12
data	Tweedie on $p_\sigma(\cdot \mid \boldsymbol{x}_1, \boldsymbol{y})$; CG prox, $\lambda = \sigma_n^2/(c\sigma^2)$	Props. 13, 14

Table 2: From the MAP objective to the bridge sampler. Code: `sb/base.py` (`reverse_sample`), `sb/immap_sb_ascent.py`, `sb/i2sb_recon.py`.

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